

THE SIMPLE DIRECTED CONNECTIVITY PROBLEM AT THREE PATHS

ABSTRACT. For a simple digraph D , let $\lambda_D(u, v)$ be the maximum number of arc-disjoint directed u - v paths. We determine the largest number of arcs in an n -vertex simple digraph satisfying $\lambda_D(u, v) \leq 2$ for all ordered pairs. The answer is

$$\max \left\{ 3(n-1), \left\lfloor \frac{(n+1)^2}{4} \right\rfloor \right\}.$$

The proof is human except for an exact finite verification through order 10. Its main ingredients are a safe directed splitting operation and a sharp counting lemma for posets in which every interval has at most three elements.

1. STATEMENT AND CONSTRUCTIONS

For $n \geq 3$, write

$$g(n) = \max\{|A(D)| : |V(D)| = n, \lambda^{\max}(D) \leq 2\}$$

and put

$$Q(n) = \left\lfloor \frac{(n+1)^2}{4} \right\rfloor, \quad F(n) = \max\{3(n-1), Q(n)\}.$$

A digraph is *simple* here if it has no loops and at most one arc in each direction on an unordered pair; digons are allowed.

Theorem 1.1. *For every $n \geq 3$,*

$$g(n) = F(n) = \max \left\{ 3(n-1), \left\lfloor \frac{(n+1)^2}{4} \right\rfloor \right\}.$$

Both lower bounds are immediate. For the linear one, take a hub joined in both directions to all other vertices and put a directed cycle on the $n-1$ non-hub vertices. There are $3(n-1)$ arcs, and every ordered pair has one endpoint of in-degree or out-degree at most 2.

For the quadratic one, split the vertices into sets X, Y , put every arc from X to Y , and put a directed cycle on Y . Taking $|Y| \in \{\lfloor (n+1)/2 \rfloor, \lceil (n+1)/2 \rceil\}$ gives

$$|Y|(|X| + 1) = Q(n).$$

A path ending at $y \in Y$ can use only the direct arc from its initial vertex in X or the unique cycle arc entering y ; hence the maximum local arc-connectivity is at most 2.

It remains to prove the upper bound. Notice that $Q(n) > 3(n-1)$ from $n = 9$ onward. Thus the induction below concerns the quadratic branch, with the finite seam $3 \leq n \leq 10$ treated exactly in Section 6.

2. SAFE SPLITTING

Definition 2.1. Let v be a vertex of a simple digraph D . A pair

$$x \rightarrow v, \quad v \rightarrow y$$

is *legal* if $x \neq y$ and $x \rightarrow y \notin A(D)$. Splitting this pair means deleting the two displayed arcs and adding $x \rightarrow y$.

The *split graph* at v is the bipartite graph with left class $N^-(v)$, right class $N^+(v)$, and an edge xy precisely when the pair $x \rightarrow v, v \rightarrow y$ is legal. Let $\nu(v)$ be its matching number.

Lemma 2.2 (Safe splitting). *Let M be a matching of size r in the split graph at v . Split all pairs represented by M and then delete v and all its remaining incident arcs. The resulting simple digraph D' satisfies*

$$|A(D')| = |A(D)| - d(v) + r, \quad \lambda^{\max}(D') \leq \lambda^{\max}(D).$$

Proof. The matching condition gives distinct incoming arcs and distinct outgoing arcs at v . The new arcs are pairwise distinct, are not loops, and were absent before, so D' is simple.

Take arc-disjoint directed paths in D' . Replace every new arc $x \rightarrow y$ by the two-arc passage $x \rightarrow v \rightarrow y$ from which it arose. Because the split pairs form a matching, the lifted directed walks are still arc-disjoint. Erasing directed closed subwalks turns each of them into a path without introducing an arc intersection. Thus every family of arc-disjoint paths in D' lifts to one of the same size in D . The arc count is immediate. $\square$

The following consequence is the reason for introducing the operation.

Corollary 2.3. *Suppose Theorem 1.1 is known for $n-1$, and let D be an n -vertex counterexample with exactly $F(n) + 1$ arcs. Then every vertex v satisfies*

$$\nu(v) \leq d(v) - (F(n) + 1 - F(n-1)).$$

In particular, equality $d(v) = F(n) + 1 - F(n-1)$ forces $\nu(v) = 0$; equivalently,

$$(*) \quad x \rightarrow v, v \rightarrow y, x \neq y \implies x \rightarrow y.$$

Proof. Apply Lemma 2.2 to a maximum matching and then use the $(n-1)$ -vertex upper bound. $\square$

We call a vertex satisfying $(*)$ *transitive*.

3. A POSET LEMMA

The low-degree vertices in a minimal even counterexample will all be transitive. Their induced digraph is therefore the comparability digraph of a poset of height at most three, with a stronger interval restriction. We isolate the exact estimate needed later.

Lemma 3.1 (Rectangle packing). *Let A, C be sets of sizes a, c , and let $X_i \times Y_i \subseteq A \times C$ ($1 \leq i \leq b$) be nonempty, pairwise edge-disjoint rectangles. Then*

$$\sum_{i=1}^b (|X_i| + |Y_i|) \leq \max\{a + bc, c + ab\}.$$

If $c \geq a$, the right-hand side is $a + bc$.

Proof. Assume $c \geq a$ and put $x_i = |X_i|$, $y_i = |Y_i|$. Edge-disjointness gives $\sum_i x_i y_i \leq ac$. Hence

$$c\left(\sum_i x_i - a\right) \leq \sum_i x_i(c - y_i) \leq c \sum_i (c - y_i).$$

After cancelling c , this is $\sum_i (x_i + y_i) \leq a + bc$. The other inequality follows by symmetry. $\square$

Lemma 3.2 (Posets with short intervals). *Let P be a finite poset on N non-isolated elements such that every interval contains at most three elements. Let A be its minimal elements, C its maximal elements, and B the remaining elements, with respective sizes a, c, b . Then B is an antichain. For $z \in B$, put*

$$X_z = \{x \in A : x < z\}, \quad Y_z = \{y \in C : z < y\}.$$

The rectangles $X_z \times Y_z$ are nonempty and pairwise edge-disjoint. After replacing P by its dual if necessary, we may assume $c \geq a$, and then

$$(1) \quad e(P) \leq c(N - c) + a.$$

Moreover, the following near-equality statements hold. Write

$$\delta = c(N - c) + a - e(P).$$

- (i) *If $\delta = 0$ and $c > a$, then all relations from A to C are present, every $Y_z = C$, and the sets X_z partition A .*
- (ii) *If $\delta = 0$, $c = a$, and $b \geq 2$, the same conclusion holds after possibly dualising P .*
- (iii) *Suppose $b = 2$, $\delta = 1$, and $c - a \geq 2$. Then all relations from A to C are present, the two sets X_1, X_2 are disjoint, and precisely one of the following occurs:*
 - (a) *$Y_1 = Y_2 = C$ and $A \setminus (X_1 \cup X_2)$ has one element;*
 - (b) *$X_1 \cup X_2 = A$, one of Y_1, Y_2 equals C , and the other misses one element of C .*

The same conclusion, after duality, holds when $c = a$.

(iv) If $b = 1$, with middle element z , set

$$\alpha = a - |X_z|, \quad \beta = c - |Y_z|, \quad \gamma = ac - e(A, C).$$

Then $\delta = \alpha + \beta + \gamma$; if $\delta > 0$, then $\alpha + \beta \geq 1$.

Proof. If $z < z'$ with $z, z' \in B$, choose $x < z$ minimal and $y > z'$ maximal. Then $[x, y]$ has at least four elements, impossible. Thus B is an antichain. Every X_z and Y_z is nonempty. If $(x, y) \in (X_z \times Y_z) \cap (X_{z'} \times Y_{z'})$ for distinct z, z' , then $[x, y]$ again contains four elements. The rectangles are therefore edge-disjoint.

Every comparability is either from A to C , from A to B , or from B to C , so

$$e(P) = e(A, C) + \sum_{z \in B} (|X_z| + |Y_z|).$$

Equation (1) now follows from $e(A, C) \leq ac$ and Lemma 3.1.

For the refinements, put

$$\gamma = ac - e(A, C), \quad \rho = a + bc - \sum_{z \in B} (|X_z| + |Y_z|).$$

Then $\delta = \gamma + \rho$, with $\gamma, \rho \geq 0$. Equality in the proof of Lemma 3.1 shows that if $\rho = 0$ and $c > a$, then every $Y_z = C$ and the X_z partition A . When $c = a$, rectangles of the two possible equality types cannot coexist, so duality gives the same conclusion. This proves (i) and (ii).

For (iii), if $X_1 \cap X_2 \neq \emptyset$, then $Y_1 \cap Y_2 = \emptyset$, and consequently

$$\sum_{i=1}^2 (|X_i| + |Y_i|) \leq 2a + c,$$

which is at least $c - a \geq 2$ below $a + 2c$. Hence X_1, X_2 are disjoint. Now

$$\rho = (a - |X_1| - |X_2|) + (c - |Y_1|) + (c - |Y_2|).$$

The case $\rho = 0$ would force the rectangles to cover $A \times C$, and then $\gamma = 0$; thus $\delta = 1$ forces $\rho = 1$ and $\gamma = 0$. The two displayed alternatives are exactly the two ways in which three nonnegative integers can sum to one. If $c = a$, dualise first when the X_i are not disjoint.

Finally, when $b = 1$, the identity in (iv) is immediate. If $\alpha = \beta = 0$, then $X_z \times Y_z = A \times C$ forces $e(A, C) = ac$, so also $\gamma = 0$. $\square$

For orientation arguments we use one more elementary obstruction.

Lemma 3.3 (Saturated rectangle obstruction). *Let L be a set of transitive vertices in a simple digraph with $\lambda^{\max} \leq 2$, and let $A_0, C_0 \subseteq L$. Suppose that for every $a \in A_0$ there is $z(a) \in L$ such that, for every $c \in C_0$,*

$$(**) \quad a \rightarrow c, \quad a \rightarrow z(a), \quad z(a) \rightarrow c.$$

Let $H = V(D) \setminus L$. Then for every $h \in H$,

$$h \not\rightarrow a, \quad c \not\rightarrow h, \quad \text{and} \quad \{a \rightarrow h, h \rightarrow c\} \not\subseteq A(D)$$

for all $a \in A_0$, $c \in C_0$. In particular, if all arcs between H and a, c are incident in the forced directions, then

$$(2) \quad d_H(a) + d_H(c) \leq |H|.$$

Proof. If $h \rightarrow a$, transitivity at a and $(**)$ give $h \rightarrow z(a)$ and $h \rightarrow c$. The direct path $h \rightarrow c$ and the paths through a and $z(a)$ are three arc-disjoint h - c paths. Thus $h \not\rightarrow a$. The assertion $c \not\rightarrow h$ follows dually, using transitivity at c . Finally, $a \rightarrow h \rightarrow c$ cannot coexist with the direct path and the path through $z(a)$. Therefore the out-neighbours of a in H and the in-neighbours of c in H are disjoint, proving (2). $\square$

4. INDUCTION: THE ODD STEP

We now prove the upper bound, assuming the finite verification of Section 6.

Lemma 4.1 (Odd step). *Let $k \geq 5$. If $g(2k) = k(k+1)$, then*

$$g(2k+1) = (k+1)^2.$$

Proof. The construction gives the lower bound. Suppose an $(2k+1)$ -vertex feasible digraph has $(k+1)^2 + 1$ arcs. Deleting any vertex leaves at most $k(k+1)$ arcs, so every vertex has degree at least $k+2$. But

$$\frac{2((k+1)^2 + 1)}{2k+1} < k+2$$

for $k \geq 5$, contradicting the average-degree bound. $\square$

5. INDUCTION: THE EVEN STEP

Lemma 5.1 (Even step). *Let $k \geq 6$. If Theorem 1.1 holds at every order below $2k$, then*

$$g(2k) = k(k+1).$$

Proof. Suppose that a feasible digraph D on $2k$ vertices has

$$E = k(k+1) + 1$$

arcs. Since $g(2k-1) = k^2$, deletion gives $d(v) \geq k+1$ for every vertex. As

$$\sum_v d(v) = 2E = 2k(k+1) + 2,$$

exactly one of the following degree patterns occurs:

- (A) $(k+1)^{2k-1}, k+3$;
- (B) $(k+1)^{2k-2}, (k+2)^2$.

Let L be the vertices of degree $k+1$ and let $H = V(D) \setminus L$. Corollary 2.3 gives $\nu(v) = 0$ for every $v \in L$, so all vertices of L are transitive.

There is no digon inside L . Indeed, deleting its two endpoints would leave

$$E - 2(k+1) + 2 = k(k-1) + 1$$

arcs on $2k - 2$ vertices, whereas the induction hypothesis gives $g(2k - 2) = k(k - 1)$. Thus $D[L]$ is the comparability digraph of a poset P . Every interval of P has at most three elements: two distinct strict intermediate elements, together with the direct arc, would give three arc-disjoint paths. No vertex of P is isolated, since a vertex of L can be incident with at most $2|H| \leq 4$ arcs outside L , while $k + 1 \geq 7$.

Let A, B, C be the minimal, middle, and maximal levels from Lemma 3.2, with sizes a, b, c , after dualising so that $c \geq a$. The set B is nonempty. Indeed, if $b = 0$, then $e(P) \leq \lfloor |L|^2/4 \rfloor$, which is smaller than the value of $e(P)$ in either degree pattern below.

Pattern (A): one exceptional vertex. Here $|H| = 1$ and

$$N = |L| = 2k - 1, \quad e(P) = E - (k + 3) = k^2 - 2.$$

Write

$$c = k - 1 + s, \quad a = k - s - b.$$

Lemma 3.2 gives

$$k^2 - 2 \leq c(N - c) + a = k^2 - s^2 - b,$$

so $s^2 + b \leq 2$. The condition $c \geq a$ leaves exactly

$$(s, b) = (1, 1), (0, 1), (0, 2).$$

For $(1, 1)$, the deficit in Lemma 3.2 is zero and the unique middle vertex has internal degree $a + c = 2k - 2 > k + 1$, impossible. For $(0, 1)$, the deficit is one; part (iv) of that lemma shows that the middle vertex has internal degree at least $a + c - 1 = 2k - 3 > k + 1$.

It remains that $(s, b) = (0, 2)$. Then $a = k - 2$, $c = k - 1$, and equality holds in Lemma 3.2. Hence all $A \rightarrow C$ relations are present, $Y_1 = Y_2 = C$, and X_1, X_2 partition A . Each middle vertex has internal degree $c + |X_i| \leq k + 1$, so $|X_i| \leq 2$. Therefore $k = 6$ and $|X_1| = |X_2| = 2$. Every vertex of A and C then has internal degree 6 and exactly one incident arc to the exceptional vertex, whereas the middle vertices have no external arc. Conditions $(**)$ hold with $A_0 = A$, $C_0 = C$ and the unique middle vertex assigned to each a . Equation (2) gives $1 + 1 \leq 1$, a contradiction.

Pattern (B): two exceptional vertices. Let $r \in \{0, 1, 2\}$ be the number of arcs between the two vertices of H . Then

$$N = |L| = 2k - 2, \quad e(P) = E - (2(k + 2) - r) = k(k - 1) - 3 + r.$$

Write

$$c = k - 1 + s, \quad a = k - 1 - s - b.$$

The poset bound becomes

$$(3) \quad s^2 + s + b \leq 3 - r.$$

Together with $c \geq a$, this gives precisely the cases in Table 1. The symbol “degree” means that one middle vertex already has internal degree greater than $k + 1$. The symbol (p, q) means that Lemma 3.3 applies to nonempty

sets A_0, C_0 whose vertices have respectively p and q incident arcs to H ; then (2) says $p + q \leq 2$, a contradiction.

r	(s, b)	(a, c)	δ	obstruction
2	(0, 1)	$(k - 2, k - 1)$	0	degree
1	(0, 1)	$(k - 2, k - 1)$	1	degree
1	(0, 2)	$(k - 3, k - 1)$	0	(1, 2)
1	(-1, 2)	$(k - 2, k - 2)$	0	(2, 1)
0	(1, 1)	$(k - 3, k)$	0	degree
0	(0, 3)	$(k - 4, k - 1)$	0	(1, 2)
0	(-1, 3)	$(k - 3, k - 2)$	0	(2, 1)
0	(0, 2)	$(k - 3, k - 1)$	1	(1, 2)
0	(-1, 2)	$(k - 2, k - 2)$	1	(2, 1)
0	(0, 1)	$(k - 2, k - 1)$	2	degree, except $k = 6$

TABLE 1. All integer solutions of (3).

We justify the table entries. In each zero-deficit row with $b \geq 2$, parts (i)–(ii) of Lemma 3.2 give, after duality when needed, all $A \rightarrow C$ relations, $Y_z = C$, and a partition of A by the sets X_z . Consequently an A -vertex has internal degree $c + 1$ and a C -vertex has internal degree $a + b$. Their external degrees are exactly the pair shown in the final column, and Lemma 3.3 applies.

The two one-deficit rows with $b = 2$ follow from part (iii) of Lemma 3.2. If one A -vertex is unused, take $A_0 = X_1 \cup X_2$ and $C_0 = C$. If one middle vertex misses one maximal element, take $A_0 = A$ and delete that maximal element from C_0 . The same external degrees, (1, 2) or (2, 1) after duality, hold on these sets.

For the last row, let z be the unique middle element and use the notation α, β, γ from Lemma 3.2(iv). Its internal degree is

$$a + c - (\alpha + \beta) \geq 2k - 5.$$

This exceeds $k + 1$ when $k > 6$. If $k = 6$, equality forces $\alpha + \beta = 2$ and $\gamma = 0$. Put $A_0 = X_z$, $C_0 = Y_z$. Every $a \in A_0$ has internal degree 6 and external degree 1, while every $c \in C_0$ has internal degree 5 and external degree 2. Lemma 3.3 again gives $1 + 2 \leq 2$, impossible.

All cases lead to a contradiction, proving the even step. $\square$

Proof of Theorem 1.1. The lower bounds were given in Section 1. The finite verification in Section 6 proves the theorem for $3 \leq n \leq 10$. For $n \geq 11$, use strong induction. Lemma 4.1 proves every odd order from the preceding even order, and Lemma 5.1 proves every even order at least 12 from the smaller orders. $\square$

6. THE FINITE VERIFICATION

For completeness, we describe the exact finite calculation on which the proof rests. It checks that no feasible digraph has $F(n)+1$ arcs for $3 \leq n \leq 10$; deleting arcs shows that this also excludes every larger arc count.

There is one Boolean variable a_{uv} for every possible arc $u \rightarrow v$. For each ordered pair (s, t) , independent Boolean variables x_u^{st} choose a cut, with $x_s^{st} = 1$ and $x_t^{st} = 0$, and the model imposes

$$(4) \quad \sum_{\substack{u \neq v \\ x_u^{st}=1, x_v^{st}=0}} a_{uv} \leq 2.$$

By directed max-flow/min-cut, the existence of such a cut for every (s, t) is exactly the condition $\lambda^{\max} \leq 2$; this is not a relaxation.

For a target $K = F(n)+1$, deletion of one vertex gives $d(v) \geq K - F(n-1)$. The program enumerates all nondecreasing degree sequences with this lower bound and sum $2K$. It also inserts the valid splitting inequalities from Corollary 2.3: a vertex of degree d cannot have a legal-pair matching of size $d - (K - F(n-1)) + 1$. The remaining symmetry is broken by ordering outdegrees inside equal-degree blocks. At $n = 9$, reversing every arc if necessary lets one additionally assume that the first degree-5 vertex has outdegree 0, 1, or 2, and these three cases are checked separately.

n	forbidden target $F(n) + 1$	degree sequences	verdict
7	19	36	all UNSAT
8	22	68	all UNSAT
9	26	15 (45 reversal branches)	all UNSAT
10	31	2	all UNSAT

TABLE 2. The nontrivial finite checks. Orders 3–6 are immediate exhaustive cases.

The accompanying file `verify_m3_bases.py` writes the exact SMT-LIB instances and invokes Z3. The source and the branch logs are part of the supplementary material. Thus the finite input to the induction is fully reproducible. The present verification records exact UNSAT solver verdicts; it does not claim an independently replayable DRAT/LRAT proof certificate.